Under Task 1: SQL-Based Traffic Analysis

1.2 Analyse annual traffic trends

	year	total_volume	absolute_change	percentage_change
1	2012	8208767	NULL	NULL
2	2013	28177412	19968645	243.26
3	2014	15731289	-12446123	-44.17
4	2015	14181206	-1550083	-9.85
5	2016	29494821	15313615	107.99
6	2017	35428156	5933335	20.12

Meaningful Observations & Explanations

- As compared between year 2012 and 2017, it appears that there is a massive jump in traffic volumes in 2013 versus 2012 with the difference of 19,968,645 and in terms of %: a 243.26% increase which is very strange. If we dive in the dataset, for 2012, the dataset only starts recording in October to December, hence giving us only a 3-month value as compared to a full year in 2013.

	holiday	temp	rain_1h	snow_1h	clouds_all	weather_main	weather_description	date_time	traffic_volume
	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter	Filter
1	None	288.28	0.0	0.0	40	Clouds	scattered clouds	2012-10-02 09:00:00	5545
2	None	289.36	0.0	0.0	75	Clouds	broken clouds	2012-10-02 10:00:00	4516
3	None	289.58	0.0	0.0	90	Clouds	overcast clouds	2012-10-02 11:00:00	4767
4	None	290.13	0.0	0.0	90	Clouds	overcast clouds	2012-10-02 12:00:00	5026
5	None	291.14	0.0	0.0	75	Clouds	broken clouds	2012-10-02 13:00:00	4918

- In 2013, the volume is around 28.1 million, in 2016 around 29.4 million and in 2017 around 35.4 million. However in 2014 and 2015, the volume is only 15.7 and 14.1 million respectively, which looks very strange. If we dive in the dataset, it appears that the dataset for 2014 and 2015 is incomplete with 2014 recording until August period while 2015 start recording from June period.

15969	None	294.48	0.0	0.0	90	Clouds	overcast clouds	2014-08-07 23:00:00	1491
15970	None	293.45	0.0	0.0	40	Clouds	scattered clouds	2014-08-08 00:00:00	848
15971	None	292.38	0.0	0.0	1	Clear	sky is clear	2014-08-08 01:00:00	516
15972	None	287.5…	0.0	0.0	92	Rain	light rain	2015-06-11 20:00:00	2886
15973	None	297.5…	0.0	0.0	0	Clear	Sky is Clear	2015-06-12 18:00:00	4380
15974	None	292.0…	0.0	0.0	8	Clear	sky is clear	2015-06-12 20:00:00	3095
15975	None	292.0…	0.0	0.0	8	Clear	sky is clear	2015-06-12 21:00:00	3299
15976	None	294.84	0.0	0.0	76	Clouds	broken clouds	2015-06-13 21:00:00	3040
15977	None	293.1…	0.0	0.0	56	Clouds	broken clouds	2015-06-13 23:00:00	2051
15978	None	296.0…	0.0	0.0	88	Rain	light rain	2015-06-14 20:00:00	2909

1.3 Analyse temperature around holidays

Year	Holiday	Average Temp (deg C)	Average Traffic Volume
2015	New Year's Day	N.A. There are no datasets	N.A. There are no datasets
2016	New Year's Day	-6.12 Deg C	1815.0
2017	New Year's Day	-3.06 Deg C	2128.0
2015	Labor Day	21.87 Deg C	973.0
2016	Labor Day	20.02 Deg C	1064.0
2017	Labor Day	22.39 Deg C	1026.0

For New Year's Day, the average temperature warmed up by +3.06 deg C from year 2016 to 2017 and as a result, traffic volume surged substantially by +17.25%. If we look at the actual temperature from -6.12 deg C to -3.06 deg C, it gives humans a significant feeling of temperature change from a colder sub-zero environment to a mild winter freeze hence 'motivating' them to move around so the traffic relevance is high.

For Labor Day, from year 2015 to 2016, the average temperature dropped by -1.85 deg C and traffic volume grew by +9.35%. For year 2016 to 2017, the average temperature increases by +2.37 deg C and traffic volume decreases by -3.57%. As the temperature hover between 20.02 Deg C to 22.39 Dec C from year 2015 to 2017 which does not give humans a feeling of a significant temperature change, so most likely the traffic relevance is low and the slight shifts in traffic volume are likely driven by other factors.

<u>Under Task 2: Descriptive Statistics and Correlation</u>

2.1 Traffic volume statistics

	Mean	Median	Variance	Standard_Deviation	Range	Min_Value	Max_Value
1	3259.82	3380.0	3947533.43	1986.84	7280	0	7280

- **Central Tendency & Bimodal Traffic Flow:**
  The Median (3,380.0) sits slightly higher than the Mean (3,259.82). In normal descriptive statistics, this hints at a slight negative skew. However, given that this tracks interstate

commuter traffic, this relationship points to a strong bimodal distribution. The highway operates primarily in two extreme phases: packed, high-volume day-time rush hours and nearly empty, low-volume late-night hours. The mean and median sit directly in the valley between these two major daily states.

- **Extreme Daily Volatility**:
  The Standard Deviation (1,986.84) is exceptionally large representing over 60% of the mean value. This tells you that hourly traffic numbers on this interstate do not hover stably around the average. Instead, at any random hour, the vehicle volume is expected to swing drastically by roughly ±2,000 vehicles, demonstrating extreme volatility governed primarily by time-of-day variables.

- **Structural Throughput Saturation**:
  The physical Range (7,280) bounds the absolute operating capacity of this stretch of westbound I-94. The minimum value of 0 indicates total standstills, overnight construction closures, or brief sensor dropouts. The maximum value of 7,280 marks the absolute empirical ceiling—the point where the highway lanes hit complete saturation under peak rush hour flow.

2.2 Correlation analysis

	Total_Records	Correlation_Coefficient
1	48194	0.1323

Analysis of the Relationship

1. Direction of the Relationship

The correlation coefficient is positive ($r > 0$). This means that as temperatures increase, traffic volume generally tends to increase as well. Conversely, as temperatures drop (colder winter conditions), traffic volume shows a slight tendency to contract.

2. Strength of the Relationship

The relationship is very weak. In statistics, a correlation value between 0.10 and 0.29 is classified as a minor or weak positive association. Even though temperature has a

mathematically positive track, the value is so close to 0 that it proves temperature is not a dominant predictor of hourly traffic volume on a day-to-day basis.

Why correlation does not necessarily imply causation

A common trap in data analytics is assuming that because two variables move together, one causes the other. A correlation coefficient only measures mathematical alignment, not a cause-and-effect relationship, for several key reasons as follows:

1. **The Presence of Confounding Variables (Third-Variable Problem):**
   Traffic volume is strongly dictated by the hour of the day and human schedules (e.g., morning and afternoon rush hours). Coincidentally, temperatures naturally peak during the afternoon because of solar cycles—exactly when afternoon rush hour occurs. The sun warming the air does not cause people to drive; rather, standard business operating hours drive both human transit and afternoon heating simultaneously.

2. **Coincidence and Spurious Associations:**
   Over a massive dataset (nearly 48,000 rows), minor random alignments can create a statistically significant, non-zero correlation coefficient even when no actual physical mechanism links the hourly variations directly.

3. **Missing Contextual Nuance:**
   Temperature variations are highly seasonal. A driver might maintain their daily routine whether it is 15 Deg C or 25 Deg C in the summer. Extreme drops (like freezing blizzards) definitely deter travel, but the subtle day-to-day temperature movements captured by this weak 0.1323 coefficient reflect minor background fluctuations rather than a direct causative impact.

Under Task 3: Probability and Congestion Analysis

	P_Congestion	P_Clear_Weather	P_Congestion_AND_Clear_Weather	P_Clear_Given_Congestion	P_High_Temp_Given_Congestion	Independent_Check_Product	Odds_Ratio_Clear_vs_Cloudy
1	0.1473	0.2776	0.0366	0.2483	0.263	0.0409	0.736

Interpretations & Discussion

1. Statistical Independence Test

- Formula Check: P(Congestion ∩ Clear Weather) = 0.0366, while P(Congestion) x P(Clear Weather) = 0.0409.
- Conclusion: Because 0.0366 ≠ 0.0409, weather conditions and congestion cannot be considered strictly independent.
- Direction: The joint probability 0.0366 is slightly lower than what we would expect if they were purely independent 0.0409. This tells us that congestion and perfectly clear skies occur together less often than random chance would dictate.

2. Odds Ratio Interpretation

- The value: 0.736
- Interpretation: Because the odds ratio is less than 1.0, the odds of experiencing traffic congestion are lower in clear weather than in cloudy weather. Specifically, the odds of hitting a traffic jam when the weather is clear are about 26.4% lower (1 - 0.736) compared to when the sky is covered in clouds.

Summary of Weather vs. Congestion Relationship

Your probability analysis suggests a clear, measurable link between atmospheric conditions and traffic flow:

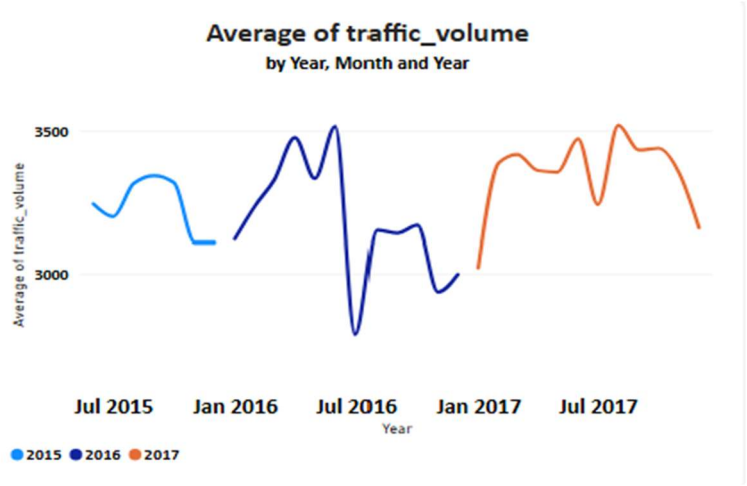
1. Weather Acts as a Commuter Buffer: When the weather turns clear, the probability of encountering clear skies given that you are already stuck in traffic drops to 24.83% (which is notably lower than the general occurrence of clear weather at 27.76%).
2. Cloudy Conditions Aggregate Risk: The odds ratio of 0.736 indicates that cloudy weather actively correlates with a higher likelihood of heavy traffic congestion. This suggests that cloudy conditions (which are frequently tied to reduced natural light, lower

temperatures, or incoming weather changes) prompt drivers to slow down or choose road travel over alternative transit modes, directly increasing the probability of hitting a breakdown or traffic bottleneck.

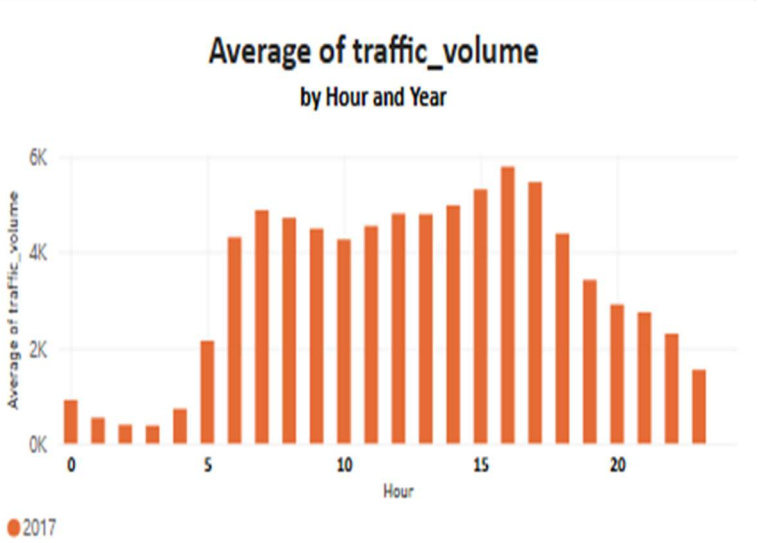
Under Task 4: Power BI Traffic Intelligence Dashboard

4.2 Dashboard analysis

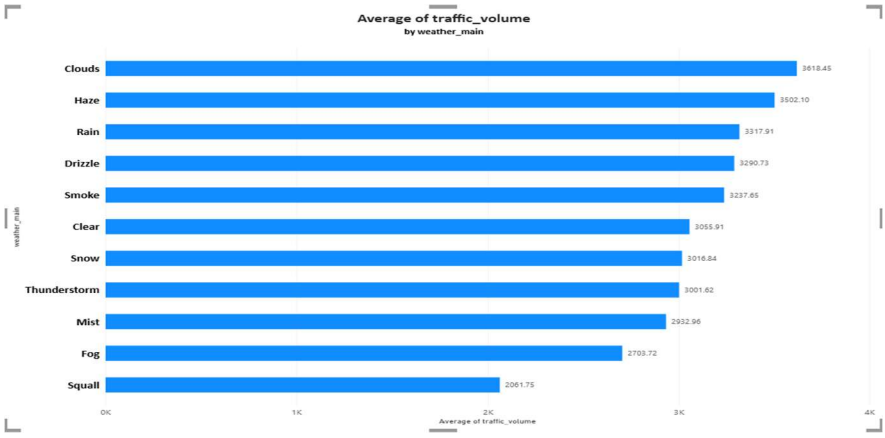
A. Daily traffic trends



B. Hourly traffic patterns

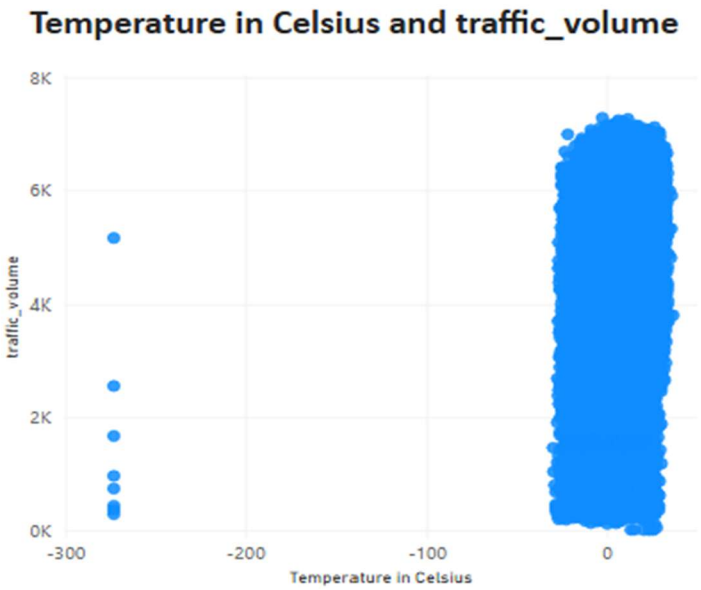


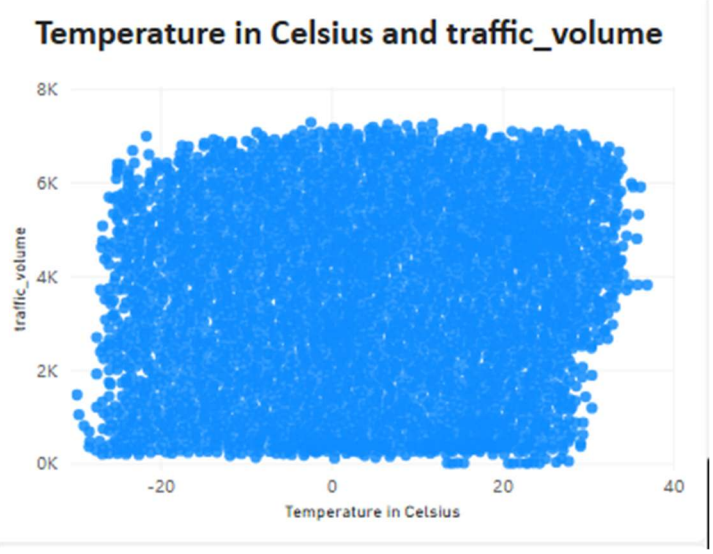
C. Weather impact



- Clouds weather conditions have the highest average traffic at 3618.45
- Squall weather conditions have the lowest average traffic at 2061.75
- The difference between the highest and lowest average traffic is 1556.70

D. Temperature and traffic





There is no visible linear relationship between temperature and traffic volume on the scatter plot. The data points do not form an upward or downward sloping line and instead form a highly dense, solid rectangular cloud in the chart. This also directly mirrors the weak Pearson Correlation coefficient of 0.1323 calculated earlier which proves that day-to-day temperature variations do not act as driving force for hourly traffic fluctuations. Traffic volume remains largely stable regardless of normal adjustments in weather temperature.

The highest traffic volume around 6000+ span incredibly wide, covering a temperature range from -15 deg C to 30 deg C. This proves the resilience of the commuters regardless of cold or warm weather.

The plot reveals a highly distinct vertical column of data points completely separated on the far left of the chart, resting at exactly -273.15 deg C. Because absolute zero is equal to -273.15 deg C, so these rows show up as extreme visual anomalies on the chart which are systemic data dropouts caused by malfunctioning highway telemetry sensors that logged a 0 value instead of a blank field.

4.3 KPI cards and filters

